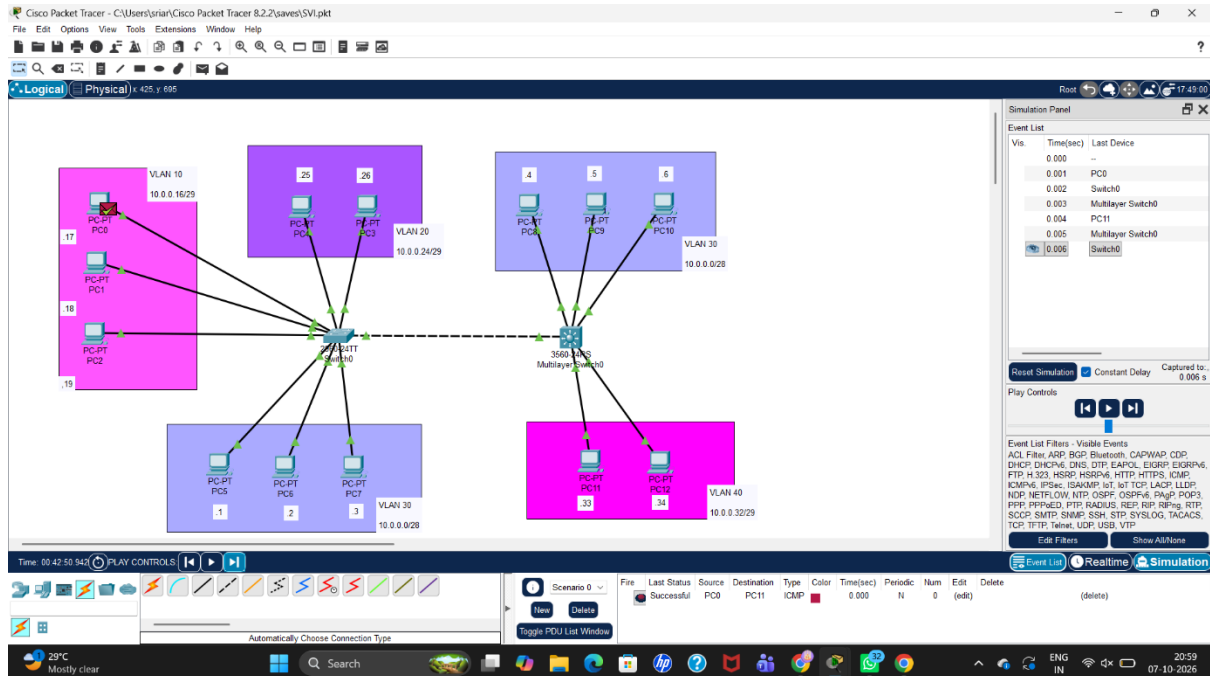


Project Title: Inter-VLAN Routing Using Switch Virtual Interface (SVI).

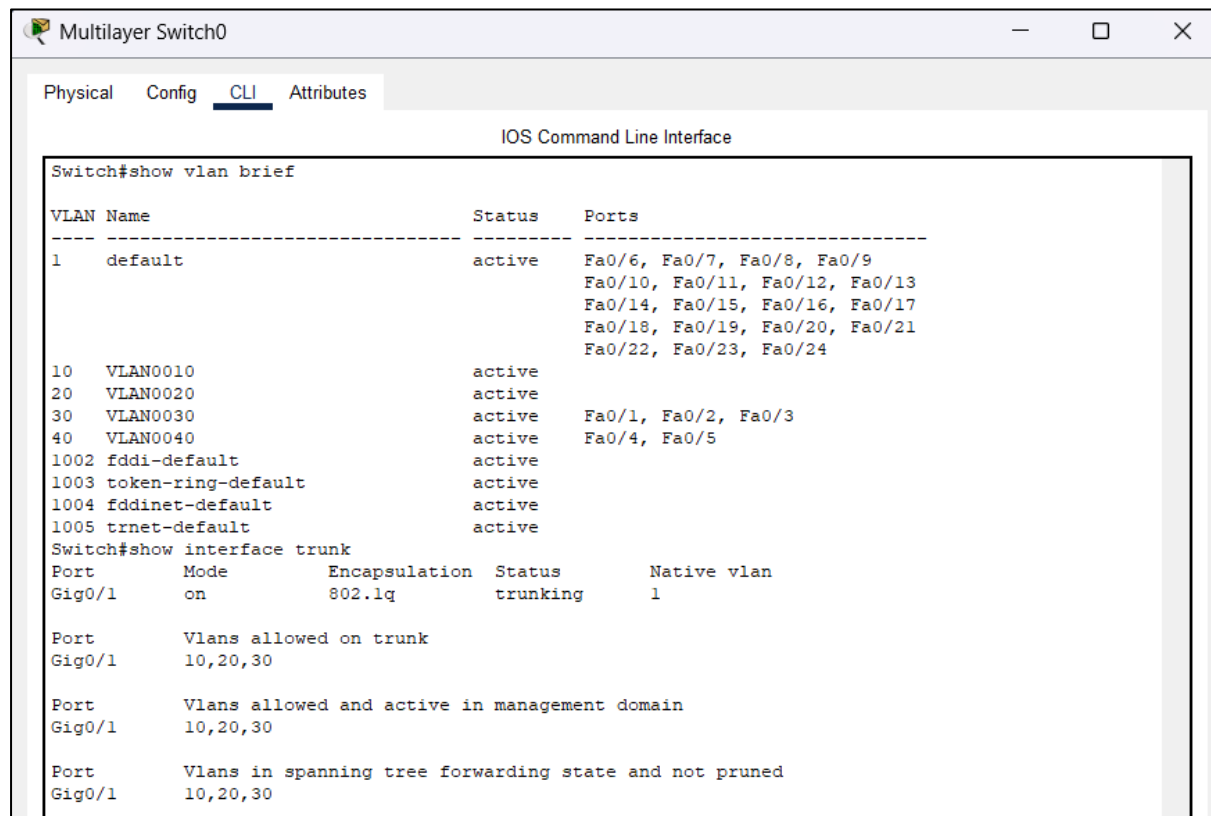
Objective: To configure multiple VLANs and enable communication between them using Switch Virtual Interfaces (SVIs) on a multilayer switch.

Network Topology:



Verification:

Device: Multilayer Switch



```

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/28 is directly connected, Vlan30
C       10.0.0.16/29 is directly connected, Vlan10
C       10.0.0.24/29 is directly connected, Vlan20
C       10.0.0.32/29 is directly connected, Vlan40
Switch#

```

Device: Switch

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```

Switch>show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/2
10   VLAN0010                active    Fa0/1, Fa0/2, Fa0/3
20   VLAN0020                active    Fa0/7, Fa0/8
30   VLAN0030                active    Fa0/4, Fa0/5, Fa0/6
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active

Switch>show interface trunk

Port      Mode          Encapsulation  Status      Native vlan
Gig0/1    on            802.1q         trunking    1

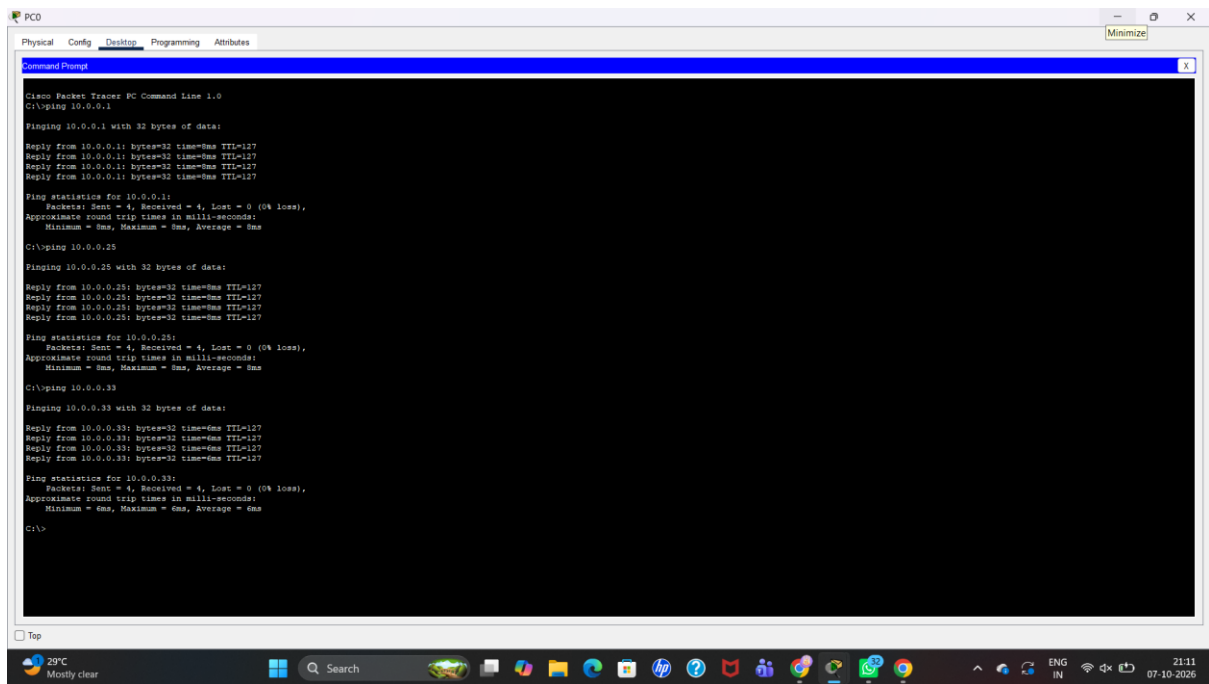
Port      Vlans allowed on trunk
Gig0/1    10,20,30

Port      Vlans allowed and active in management domain
Gig0/1    10,20,30

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    10,20,30

```

Ping test from VLAN 10 (10.0.0.17) to all other VLANs.



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:
Reply from 10.0.0.1: bytes=32 time=6ms TTL=127
Reply from 10.0.0.1: bytes=32 time=6ms TTL=127
Reply from 10.0.0.1: bytes=32 time=6ms TTL=127
Reply from 10.0.0.1: bytes=32 time=6ms TTL=127

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 6ms, Average = 6ms
C:\>ping 10.0.0.25

Pinging 10.0.0.25 with 32 bytes of data:
Reply from 10.0.0.25: bytes=32 time=6ms TTL=127
Reply from 10.0.0.25: bytes=32 time=6ms TTL=127
Reply from 10.0.0.25: bytes=32 time=6ms TTL=127
Reply from 10.0.0.25: bytes=32 time=6ms TTL=127

Ping statistics for 10.0.0.25:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 6ms, Average = 6ms
C:\>ping 10.0.0.33

Pinging 10.0.0.33 with 32 bytes of data:
Reply from 10.0.0.33: bytes=32 time=6ms TTL=127
Reply from 10.0.0.33: bytes=32 time=6ms TTL=127
Reply from 10.0.0.33: bytes=32 time=6ms TTL=127
Reply from 10.0.0.33: bytes=32 time=6ms TTL=127

Ping statistics for 10.0.0.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 6ms, Average = 6ms
C:\>
```

Top

29°C Mostly clear Search 21:11 07-10-2026